

Chan Ding Hao

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Education

Singapore Management University - Bachelor of Science in Economics

Aug 2022 - July 2026 (Expected)

- **Relevant Coursework:** Big Data Architecture, Machine Learning & Applications, Algorithms and Programming, Visual Analytics for Business Intelligence, Business Data Management, Computational Thinking & Programming, Data Analytics with R

Work Experience

Wise - Product Analyst Intern

Mar 2026 - Jul 2026

- Will be joining the Send High Volume team, focusing on analyzing high-value customer behavior and transaction patterns to identify growth opportunities

ByteDance - Payment Risk Data Analytics Intern

Jan 2026 - Mar 2026

- Conducted exploratory analysis on new user registration data to identify behavioral signals (IP geolocation, device fingerprinting, velocity patterns) indicative of spam account creation, establishing baseline fraud rates and feature candidates for a risk scoring framework
- Developed HiveSQL queries to extract and analyze 1M+ daily user records, performing quantile analysis across device and IP distributions to surface anomalous registration patterns

Razer - Data Analyst Intern (Analytics Engineering)

July 2025 - Dec 2025

- Built dashboards on Apache Superset to track key metrics (Daily CTR, Daily Revenue etc) for Razer recommendation engine using SQL, reducing reporting time by 25%.
- Built transformation models (SQL) within ELT pipelines using dbt DAGs to deliver business intelligence solutions for internal stakeholders served through Apache Superset, increasing dashboard load speed by 50%.
- Developed **deduplication logic** for user software data through **aws wrangler**, **SQL** and **pandas** in **AWS SageMaker AI** notebooks, identifying data quality issues and providing implementation specifications to data engineering.

ONE (Ocean Network Express) - Data Analyst Intern

June 2024 - Dec 2024

- Optimized truck routes using a CI/CD pipeline on GCP, leveraging Python for data transformation and analysis on VertexAI and SQL on BigQuery. Improved processing speed by 10%, enabling faster data-driven decisions and improving real-time logistics efficiency.
- Discovered important aspects related to data (e.g. Hazmat constraints), while validating discoveries with various stakeholders, reducing distance by 5%, and saving over \$500,000 USD annually with scalability potential.

Projects

Crypto Market Data Ingestion (In Progress)

Jan 2026 - April 2026 (Expected)

- Built end-to-end crypto market intelligence pipeline on AWS (EventBridge → Lambda → RDS PostgreSQL) ingesting daily BTC/USD OHLC snapshots from Kraken REST API, enabling exchange-focused business metrics analysis
- Designed and deployed Grafana dashboards tracking 5 key exchange performance metrics — trading volume, VWAP vs. close, volume-volatility correlation, 30-day volatility regime, and drawdown — to surface actionable revenue and liquidity insights

SPY Analytics and Forecasting - [Github](#) - [Medium](#) (1), (2), (3)

Aug 2024 - Dec 2024

- Conducted extensive **data wrangling, analysis, and visualisation** of SPY's historical and inflation-adjusted returns alongside Federal Reserve inflation data to highlight long-term investment advantages and performance stability.
- Applied **time-series statistical techniques**, such as **stationarity testing (ADF)** and **differencing**, to preprocess SPY data for robust time series forecasting.
- Constructed a weighted ensemble model (**LSTM -PyTorch**, **Prophet**, and **Monte Carlo simulations**) for **time series forecasting**. Predicted the log returns of SPY in the next 20 years with confidence intervals.

Skills and Certifications

Technical Skills: Python; PySpark; SQL (HiveQL, SparkSQL, MySQL, PostgreSQL); R; dbt; Bash/Linux; Git; Docker

Libraries & Frameworks: NumPy; Pandas; PyTorch; Scikit-learn; XGBoost; Plotly; Boto3; AWS Wrangler; Google Cloud SDK; Streamlit

Cloud & Platforms: GCP (BigQuery, Vertex AI); AWS (S3, Redshift, RDS, Lambda, SageMaker, Bedrock); Pinecone

BI & Visualization: Apache Superset; Grafana; Tableau

Certifications & Training [Google Data Analytics Professional Certificate](#), [HackerRank SQL \(Advanced\)](#)